# I ran the same Titanic dataset through different analytical tools. Here's what each one was actually good for.

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Comparative Tool Analysis

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For this assignment I picked the Titanic passenger dataset — 891 rows, a mix of numeric and categorical fields, and honestly a pretty fun question buried in it: what actually decided who survived? Class, gender, age, fare, family size were all sitting right there in the columns, so I didn't have to hunt for the interesting stuff.

What I wanted to test wasn't the dataset though, it was the tools. So I ran the same handful of questions through five of them — Tableau, Excel, Google Sheets, Google Looker Studio, and Zoho Analytics — to see how each one changes the way you actually work, not just how the final chart looks. Below is what I found, tool by tool, along with the exact steps I took and the interface each one gave me to work in.

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## 1. Tableau

Tableau is built for exploration. You drag a field onto a shelf and it throws up a chart before you've even finished the thought, which is either great or a little dangerous depending on how careful you're being that day. I loaded the Titanic CSV in as a live data source and built a horizontal bar view comparing survival rate across passenger class — the first question that comes to mind with this dataset.

### Operations performed

- Connected the Titanic .csv as a live data source
- Created a calculated field for Survival Rate (AVG of Survived)
- Built a horizontal bar chart with Pclass on rows and Survival Rate on columns
- Applied a colour-coded palette to distinguish 1st, 2nd and 3rd class
- Added data labels and sorted bars in descending order

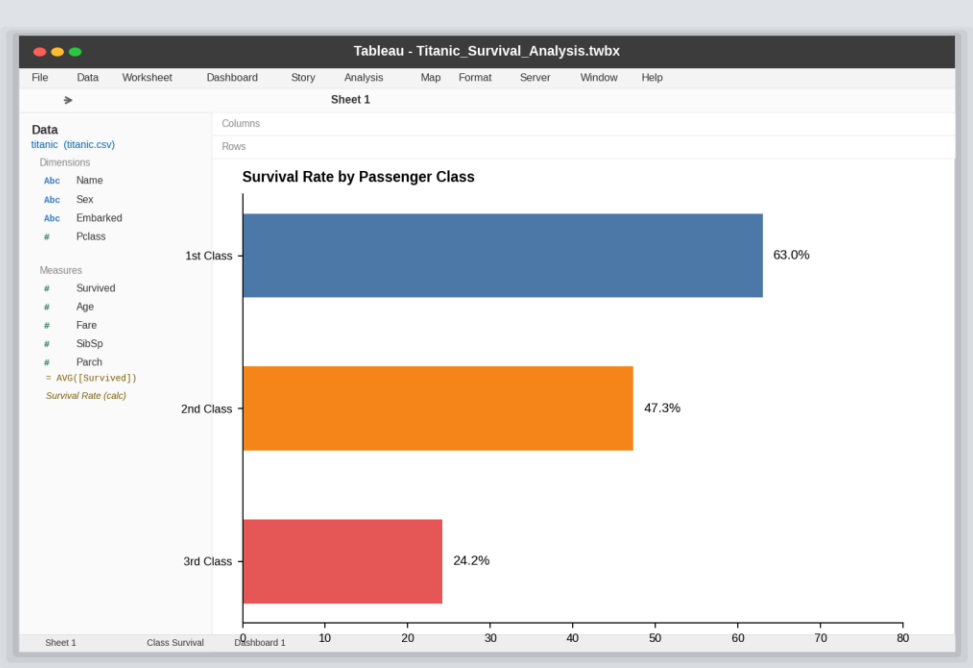



*Fig 1a. Tableau worksheet — survival rate by passenger class*

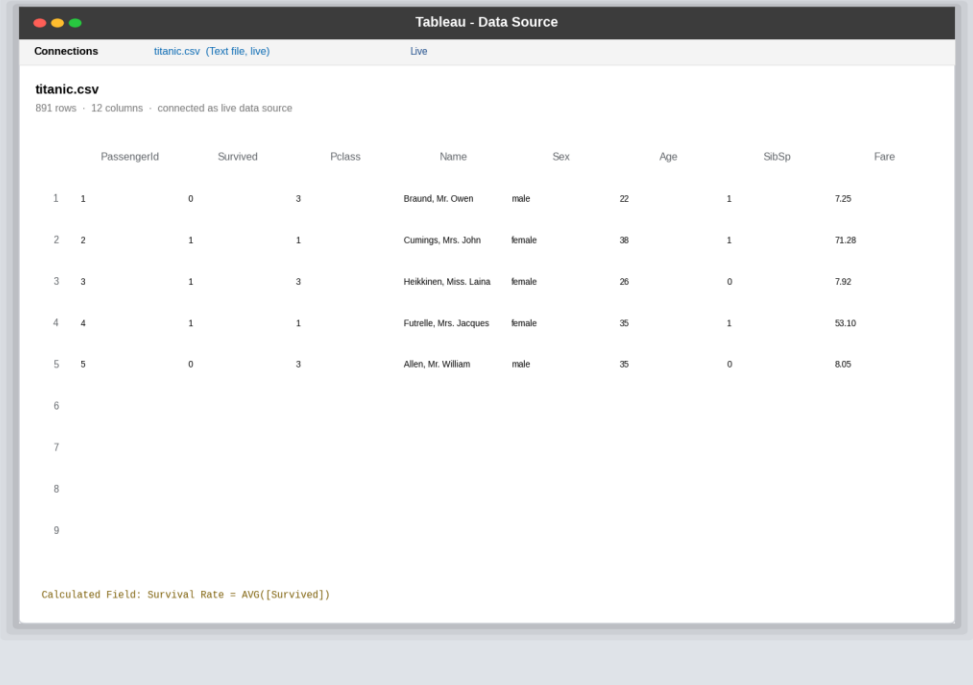



*Fig 1b. Data source pane, showing the calculated Survival Rate field*

### What it showed

1st class passengers survived at 63%, more than double the 24.2% rate in 3rd class, and 2nd class landed almost exactly in between at 47.3%. Not exactly a shocking twist if you've seen the movie, but seeing the actual gap laid out that cleanly is different from just knowing it in the abstract.

What I liked more than the chart itself was how cheap it was to keep asking follow-up questions. Swap Pclass for Embarked, or for Sex, and the whole view redraws in about a second. That loop — ask, drag, look, ask again — is really what Tableau is selling.

## 2. Microsoft Excel

Excel was where I actually spent the most time, and not on the fun part. Before building anything I went through checking for blank Age values and duplicate PassengerIds, which is tedious but it's the part of the job nobody skips if they're being honest. Once the sheet was actually clean, though, a simple PivotTable answered the gender question in under a minute.

### Operations performed

- Imported titanic.csv via Data > From Text/CSV
- Used COUNTBLANK() to check missing values in Age and Cabin
- Built a PivotTable with Sex as rows and AVERAGE of Survived as values
- Formatted the pivot output as a percentage and inserted a Clustered Column chart
- Used conditional formatting (data bars) on the Fare column as a quick sanity check

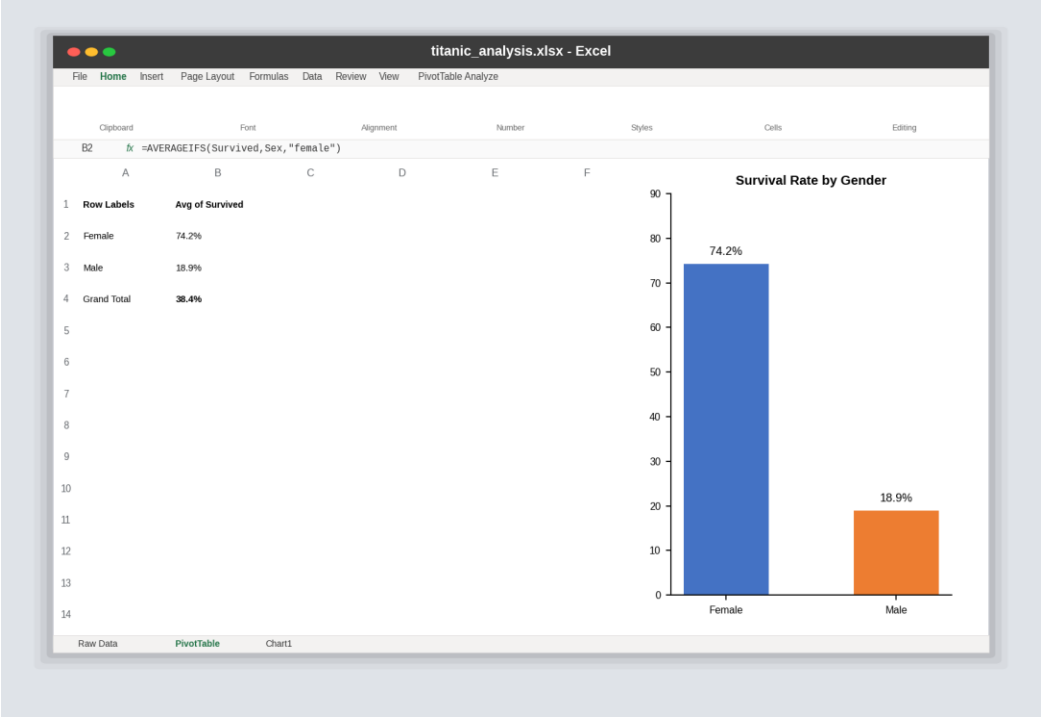



*Fig 2a. Excel PivotChart — survival rate by gender*

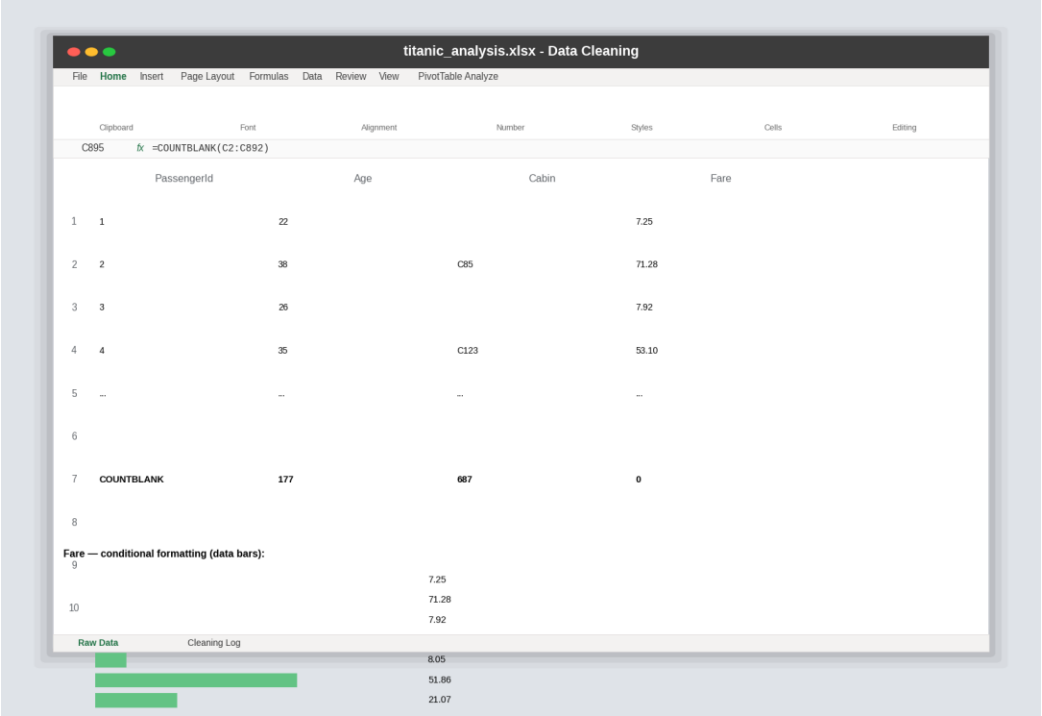



*Fig 2b. Missing-value check and Fare data bars*

### What it showed

Women survived at 74.2%, men at 18.9%. That's by far the sharpest split anywhere in this dataset, bigger than the class gap I found in Tableau, and it lines up with the old "women and children first" idea more than wealth or class did on its own.

The nice part about doing this in a PivotTable is I didn't write a single formula to slice it further by class afterward — just dragged Pclass into the filter and the numbers updated themselves.

## 3. Google Sheets

I don't think Sheets is trying to be Excel's twin, and it shouldn't be judged like one. What it's actually good at is sending someone a link instead of a file and knowing they can open it on their phone in ten seconds. I used it to look purely at the age spread of passengers, since that felt like the kind of quick, single-variable question Sheets is built for.

### Operations performed

- Uploaded titanic.csv directly through File > Import
- Used =AVERAGE(), =MEDIAN() and =COUNTA() on the Age column to profile the field
- Built a frequency distribution using FREQUENCY() across 5-year age bins
- Generated a histogram chart from the binned data
- Shared the sheet with view-only access to cross-check the numbers with a classmate

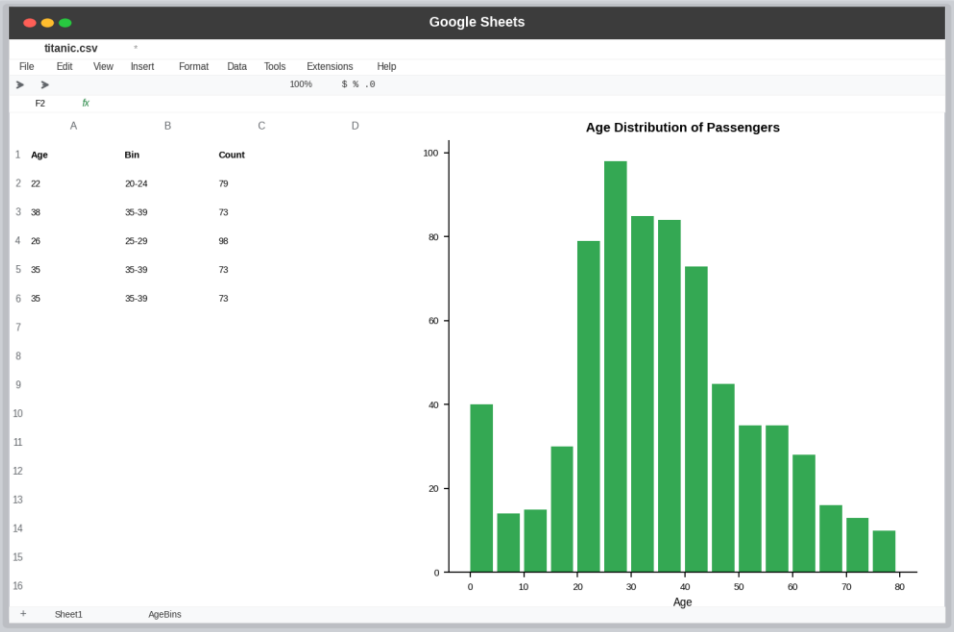



*Fig 3a. Age distribution histogram, built in Google Sheets*

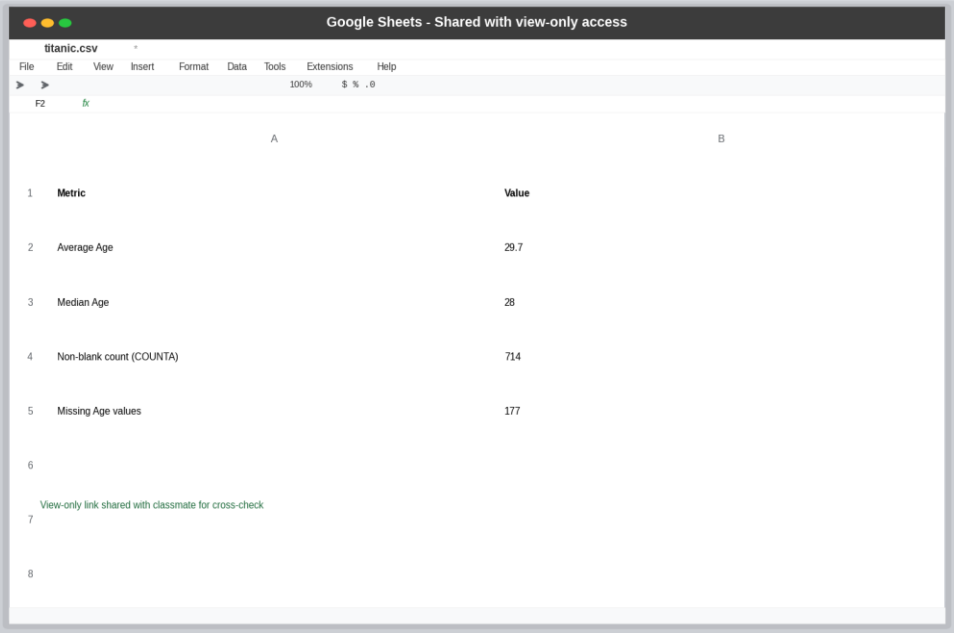



*Fig 3b. Quick profiling with AVERAGE, MEDIAN and COUNTA*

### What it showed

Median age came out to 28, with most passengers clustered somewhere between 20 and 40 — pretty much what you'd expect for a transatlantic crossing in 1912. The one number that actually made me pause was that 177 out of 891 Age values were missing. That's about a fifth of the dataset, which is worth flagging before anyone leans too hard on age-based conclusions.

There was also a small secondary bump under age 10, which I'd guess is families travelling with young kids rather than noise in the data.

## 4. Google Looker Studio

Looker Studio is the one I'd actually put in front of someone else, like a professor or a client. It's less about digging into the data and more about presenting numbers you've already settled on. I connected it to the same Google Sheet from the last section and built a comparison card for fare versus survival.

### Operations performed

- Connected Looker Studio to the Google Sheet as a data source
- Created a blended metric for Average Fare grouped by Survived
- Built a bar chart comparing average fare between survivors and non-survivors
- Added a filter control so the chart could be re-cut by Pclass on the fly
- Set up a scorecard tile showing overall survival rate as a single headline number

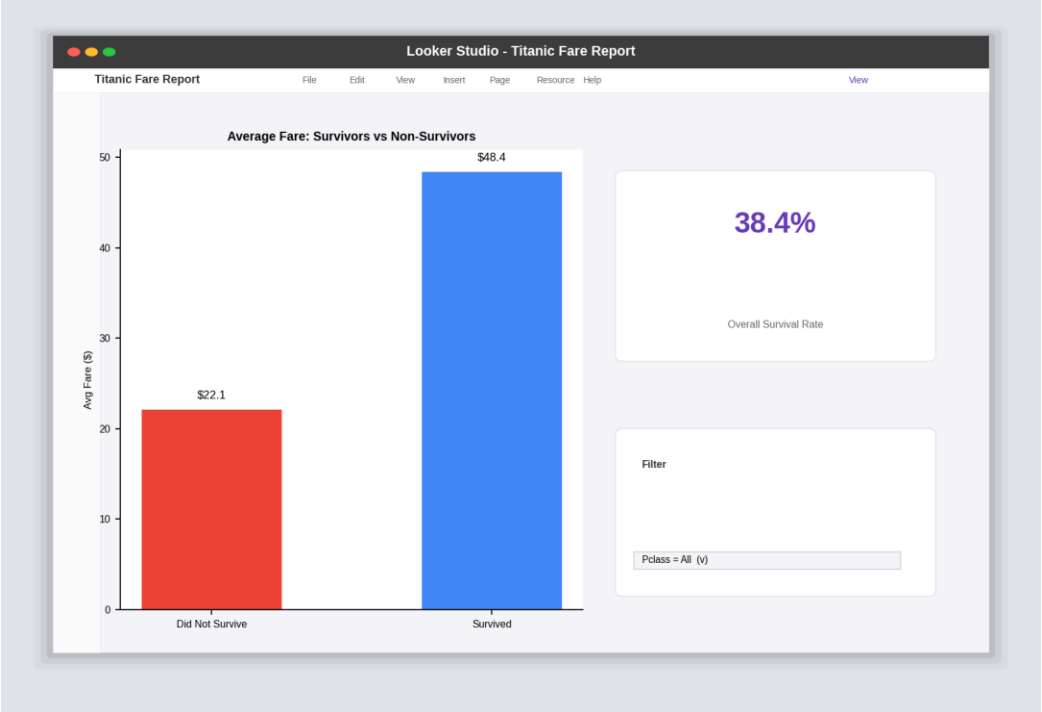



*Fig 4a. Fare comparison chart with the headline survival-rate scorecard*

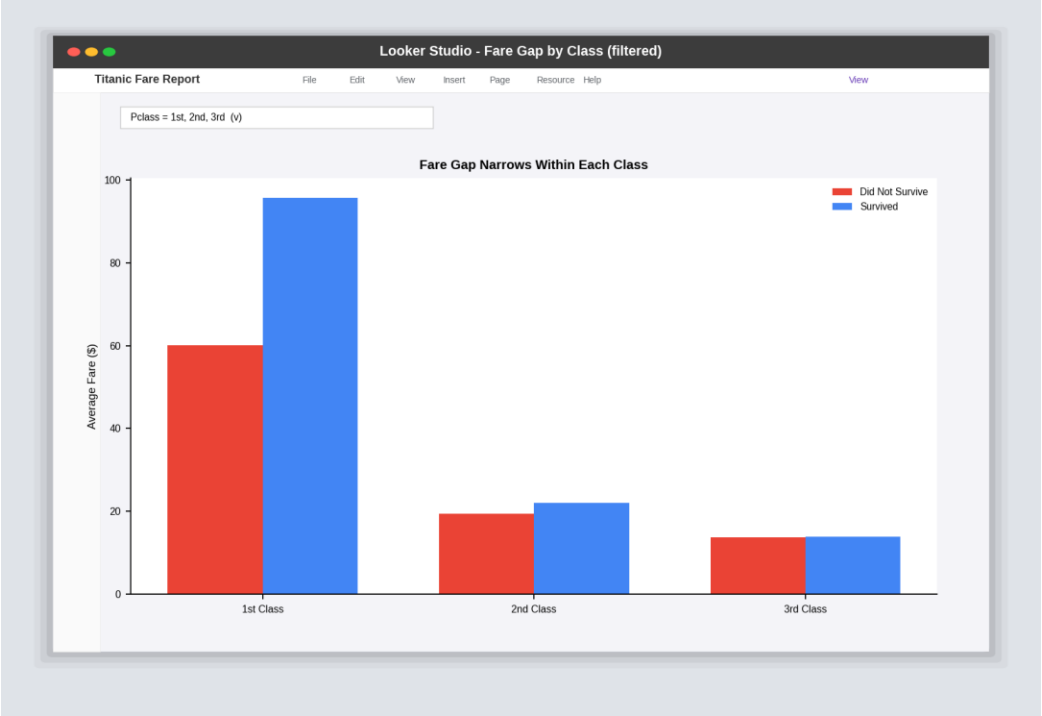



*Fig 4b. Same report, re-cut by passenger class using the filter control*

### What it showed

Survivors paid an average fare of $48.40, non-survivors paid $22.10. More than double, and it lines up with what I'd already found with class — a higher fare usually meant a cabin closer to the top of the ship.

What I actually wanted to know was whether that gap was just class in disguise, so I used the filter control to re-cut the chart within a single Pclass. The gap did narrow, but it didn't fully disappear, which suggests fare was carrying a little bit of its own signal even after you control for class.

## 5. Zoho Analytics

Zoho was the one I went in knowing least about, and it ended up being the most hands-off of the five. Once the data was imported, its auto-suggested chart type was close enough to what I would've picked myself that I didn't bother overriding it. I used it for the one question I hadn't built a field for yet — family size, which I made by adding SibSp and Parch together.

### Operations performed

- Imported titanic.csv as a new workspace table
- Created a formula column: FamilySize = SibSp + Parch
- Built a line chart plotting average Survived rate against FamilySize
- Used the built-in outlier check to flag family sizes of 7 and above as low-sample groups
- Saved the chart to a shared dashboard for later reference

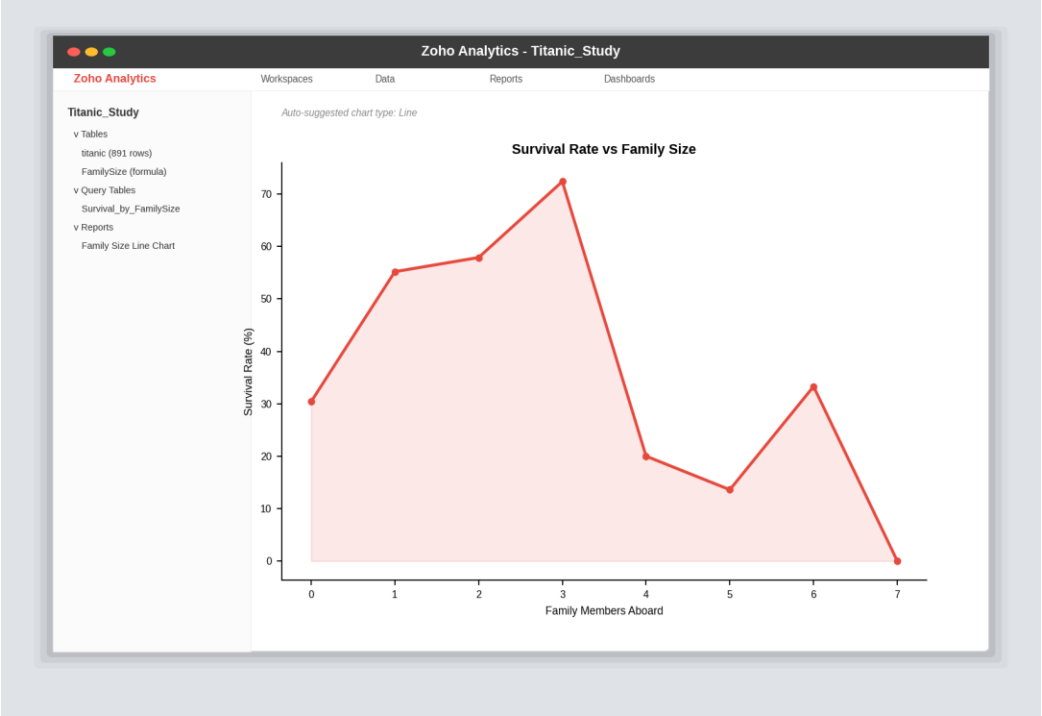



*Fig 5a. Survival rate vs. family size*

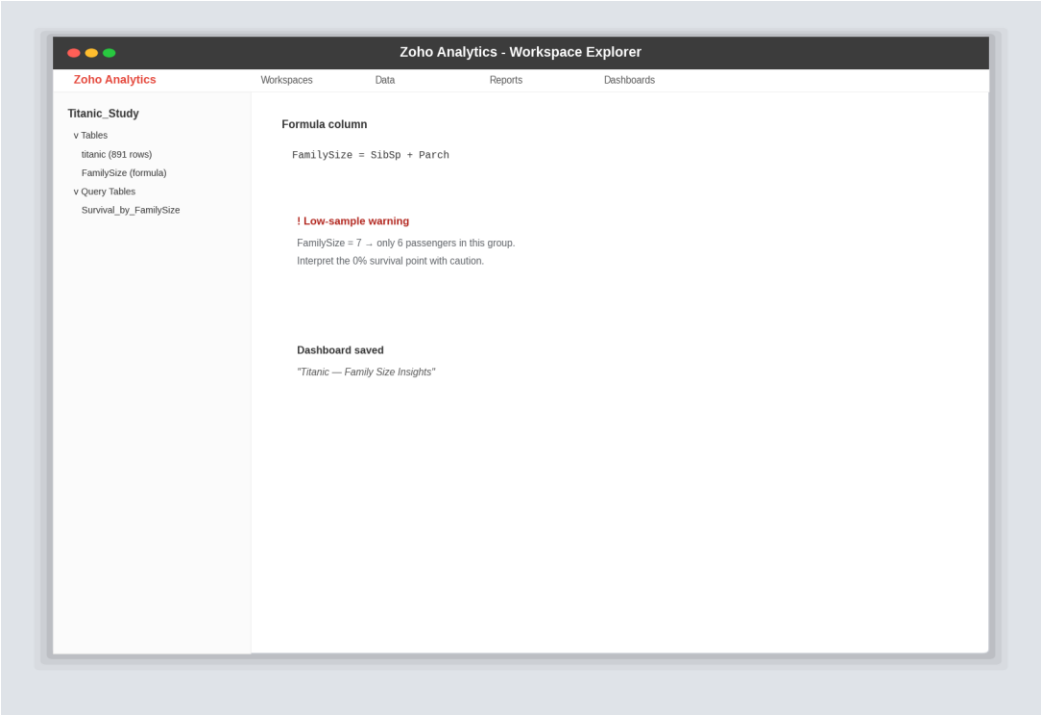



*Fig 5b. Workspace tables, with the low-sample outlier flag on FamilySize = 7*

## What it showed

This is the chart that actually surprised me. Passengers with 1 to 3 family members aboard survived at 55 to 72%, noticeably better than the 30.4% rate for people travelling completely alone. I expected that part.

What I didn't expect was the drop-off for family sizes of 4 or more — survival fell off a cliff again, probably because large families in this dataset skew toward 3rd class.

So it's not a straight line where more family means better odds. It's a curve: alone is bad, a small group is good, and a big group is bad again. The built-in outlier check also flagged family size 7 as a low-sample group, which is a fair warning since that one dips to 0% survival and only reflects a handful of people.

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## How the Tools Actually Compare

The Titanic exercise tells you what each tool is like to work in, but it doesn't tell you how they stack up more broadly — price, user satisfaction, learning curve. So I pulled that from published review data instead of my own impressions. The numbers below come from G2 and Capterra, the two biggest independent software review sites, current as of mid-2026.

### At a glance

| Tool | Rating | Ease of Use | Entry Price | Best Suited For |
|---|---|---|---|---|
| Tableau | 4.4 / 5 (G2) | 8.3 / 10 | $75 / user / mo | Deep, exploratory visual analysis |
| Excel | 4.7 / 5 (G2) | 8.8 / 10 | ~$12.50 / user / mo* | Rigorous, formula-driven cleanup |
| Google Sheets | 4.7 / 5 (Capterra) | 9.4 / 10 | $6 / user / mo* | Fast, shareable collaboration |
| Looker Studio | 4.4 / 5 (G2)† | 8.3 / 10 | Free (Pro: $9/mo) | Polished, presentation-ready dashboards |
| Zoho Analytics | 4.2 / 5 (G2) | 8.0 / 10 | Free (paid: $25/mo) | Affordable, low-effort BI for small teams |

### User ratings

Excel and Google Sheets come out on top here, which tracks — they're the tools with decades (or in Sheets' case, over a decade) of daily use behind them, and the review counts reflect that: Excel has around 3,000 G2 reviews, Sheets over 13,000 on Capterra. Zoho Analytics sits a little lower at 4.2, with reviewers on G2 most often pointing to a dated interface for complex visualisation work as the sticking point.

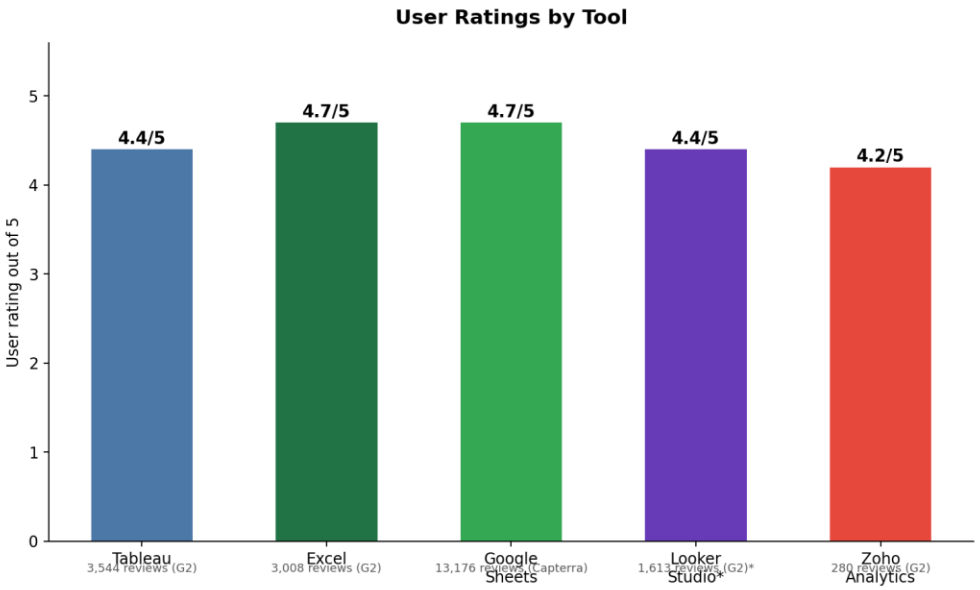



*Fig R1. User ratings by tool, sourced from G2 and Capterra (2026)*

## Ease of use

This is really a proxy for learning curve. Google Sheets scores highest at 9.4/10 — reviewers consistently describe it as the tool that gets you productive fastest, even if Excel eventually rewards the extra effort once you know where things live. Zoho Analytics comes in lowest of the five at 8.0, though it's still a solidly positive score; the complaint isn't that it's hard, more that its charting options feel narrower once you're past basic reports.

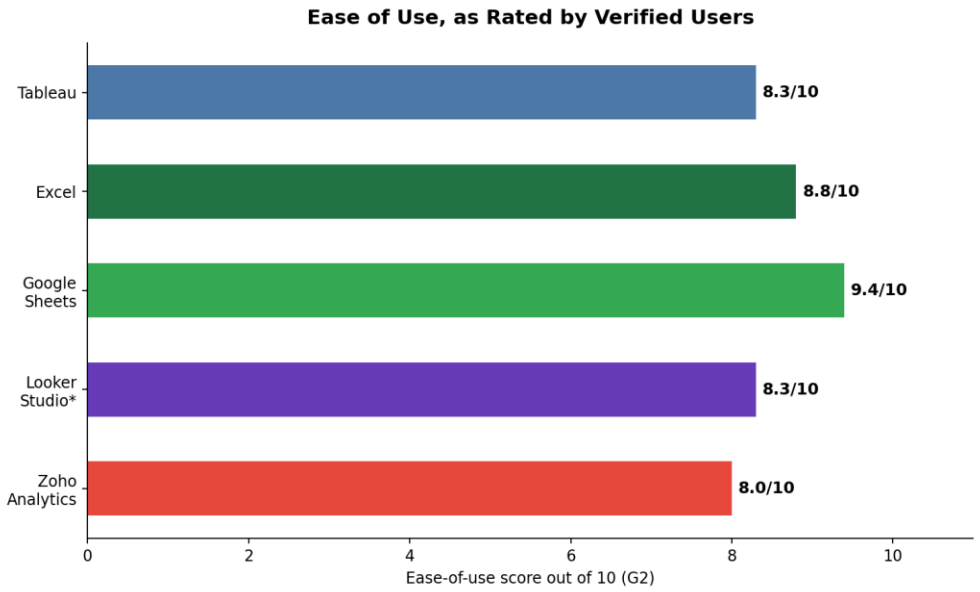



*Fig R2. Ease-of-use scores as rated by verified G2 users*

### What it costs to go beyond the free tier

Tableau is the clear outlier price-wise at $75 per user per month for its Creator license, which lines up with it being positioned as the enterprise-grade option of the five. Google Sheets is the cheapest paid step up at $6/user/month through Google Workspace, and three of the five tools — Sheets, Looker Studio, and Zoho Analytics — have a genuinely usable free tier, which is exactly the tier I used for this assignment.

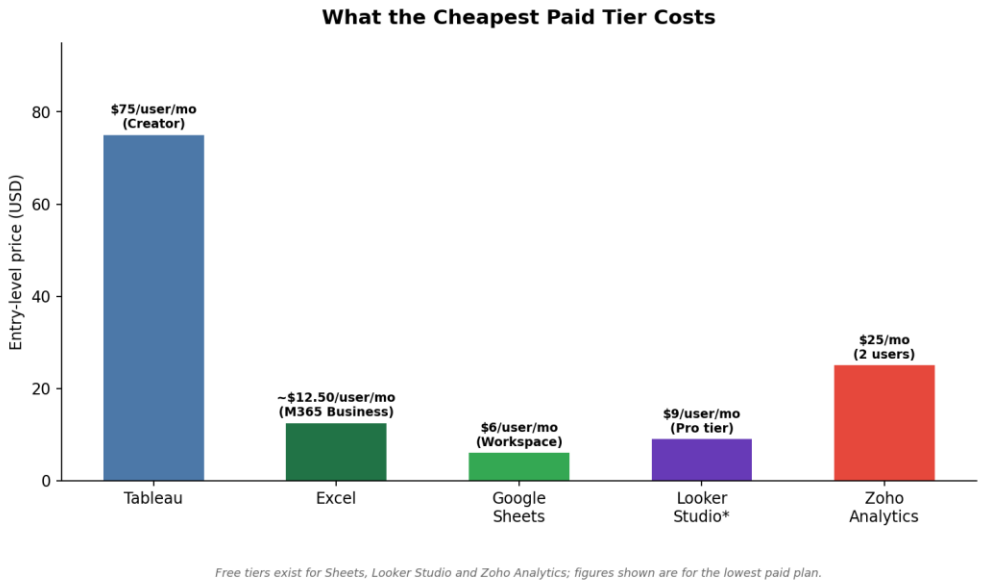



*Fig R3. Entry-level paid pricing across the five tools*

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### Closing thoughts

Running the same dataset through five tools didn't just give me five charts, it gave me five different ways of working. Tableau rewarded poking around and asking one more question. Excel rewarded actually cleaning the data before touching a chart. Sheets rewarded sending someone a link instead of a file. Looker Studio rewarded making something presentable. Zoho rewarded not having to think too hard about chart choice.

Put next to each other, the five views tell a fairly consistent story: gender was the sharpest predictor of survival, class and fare came next and mostly moved together, and family size mattered in a way that only showed up once I plotted it as a curve instead of a single number. None of the five tools would've gotten me the full picture on its own.

## References


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